

The Integral Middle Cohomology of the Theta Divisor of a Cubic Threefold

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Abstract

Let Θ be the theta divisor of the intermediate Jacobian of a smooth complex cubic threefold X , and let $M = \mathrm{Bl}_0 \Theta$. We compute the integral middle lattice of M , equivalently the degree-three intersection cohomology of Θ . It is free of rank 130 and fits into an abstractly split exact sequence with terms $\bigwedge^3 H^1(J, \mathbf{Z})$ and $H^3(X, \mathbf{Z})$. Its simultaneous restriction to X and Gysin image in J has nontrivial mod-two glue. If $L(\alpha) = \theta \wedge \alpha$, then

$$b_* H^3(M, \mathbf{Z}) = \mathrm{Sat}(L \bigwedge^3 H^1(J, \mathbf{Z})) = L \bigwedge^3 H^1(J, \mathbf{Z}) + \theta^{[2]} \wedge H^1(J, \mathbf{Z}),$$

and the saturation quotient is $(\mathbf{Z}/2)^{10}$. The proof reduces the integral Lefschetz defect to the unsigned vertex-edge incidence matrix of a complete graph. The degree-six difference map of the Fano surface then identifies the geometric glue through a Pontryagin-product formula. By duality, the degree-five quotient not pulled back from J is free of rank ten, while exceptional Gysin classes form its doubled sublattice.

Keywords. Cubic threefold; intermediate Jacobian; theta divisor; intersection cohomology; symplectic lattice; Fano surface.

1 Introduction

Let $X \subset \mathbf{P}_{\mathbf{C}}^4$ be a smooth cubic threefold and let (J, Θ) be its principally polarized intermediate Jacobian. The theta divisor has a unique singular point, an ordinary triple point at the origin, and its projectivized tangent cone is X [Bea82, Theorem 1]. Its blow-up

$$\sigma : M = \mathrm{Bl}_0 \Theta \longrightarrow \Theta$$

is smooth, with exceptional divisor X and normal bundle $\mathcal{O}_X(-1)$. Write $i : \Theta \hookrightarrow J$, $b = i\sigma$, $e : X \hookrightarrow M$, and

$$\Lambda = H^1(J, \mathbf{Z}), \quad \theta = [\Theta] \in \bigwedge^2 \Lambda.$$

The principal polarization makes Λ a unimodular symplectic lattice of rank ten. We use the divided powers $\theta^{[r]} = \theta^r / r!$, which are integral.

The ordinary group $H^3(\Theta, \mathbf{Z})$ is $\bigwedge^3 \Lambda$ by singular weak Lefschetz. Blowing up the triple point adds ten generators, but their placement relative to restriction to X and pushforward to J is not a direct sum. The following theorem describes that placement.

Define

$$L : \bigwedge^3 \Lambda \longrightarrow \bigwedge^5 \Lambda, \quad L(\alpha) = \theta \wedge \alpha,$$

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and let $S = \text{Sat}(L \bigwedge^3 \Lambda)$ inside $\bigwedge^5 \Lambda$. The cylinder correspondence of the universal line on X and the principal polarization give an integral isomorphism

$$y : H^3(X, \mathbf{Z}) \xrightarrow{\sim} \Lambda;$$

we fix its global sign in Section 4.

Theorem 1.1 (Integral middle lattice). *There is an exact sequence of free abelian groups*

$$0 \longrightarrow \bigwedge^3 \Lambda \xrightarrow{b^*} H^3(M, \mathbf{Z}) \xrightarrow{e^*} H^3(X, \mathbf{Z}) \longrightarrow 0. \quad (1)$$

It splits as a sequence of abelian groups. The Gysin image is

$$b_* H^3(M, \mathbf{Z}) = S = L \bigwedge^3 \Lambda + \theta^{[2]} \wedge \Lambda, \quad S/L \bigwedge^3 \Lambda \cong \Lambda/2\Lambda. \quad (2)$$

More precisely, (b_, e^*) embeds $H^3(M, \mathbf{Z})$ in $S \oplus H^3(X, \mathbf{Z})$, with image*

$$\left\{ (s, \gamma) : [s] = [\theta^{[2]} \wedge y(\gamma)] \text{ in } S/L \bigwedge^3 \Lambda \right\}. \quad (3)$$

In particular, $H^3(M, \mathbf{Z})$ is torsion-free of rank 130, and the nonzero Smith factors of L are $1^{110}, 2^{10}$.

The split sequence (1) alone does not see the arithmetic content: any lift of γ has push-forward in the mod-two coset $[\theta^{[2]} \wedge y(\gamma)]$. Thus “glue” below always means the simultaneous lattice (3), not a nonsplit extension of abstract abelian groups.

The dual statement corrects another tempting reading of the ten factors of two. Put

$$E_M = H^5(M, \mathbf{Z}) / (b^* H^5(J, \mathbf{Z}) + \text{tors}). \quad (4)$$

Theorem 1.2 (Escape lattice). *The group E_M is free of rank ten. The image of $e_* H^3(X, \mathbf{Z})$ in E_M is exactly $2E_M$. Consequently*

$$\frac{H^5(M, \mathbf{Z})}{b^* H^5(J, \mathbf{Z}) + e_* H^3(X, \mathbf{Z}) + \text{tors}} \cong (\mathbf{Z}/2)^{10}.$$

The lattice is intrinsic to the singular theta divisor in middle degree.

Corollary 1.3 (Integral intersection cohomology). *There are natural isomorphisms*

$$IH^3(\Theta, \mathbf{Z}) \cong H^3(M, \mathbf{Z})$$

and an exact sequence

$$0 \longrightarrow H^3(\Theta, \mathbf{Z}) \longrightarrow IH^3(\Theta, \mathbf{Z}) \longrightarrow H^3(X, \mathbf{Z}) \longrightarrow 0.$$

Under these identifications, Theorem 1.1 computes the integral lattice $IH^3(\Theta, \mathbf{Z})$.

The proof has three parts. First, the link of the triple point is the circle bundle of $\mathcal{O}_X(-1)$; its Gysin sequence and Mayer–Vietoris give (1). Second, a symplectic-basis decomposition turns $L : \bigwedge^3 \Lambda \rightarrow \bigwedge^5 \Lambda$ into primitive blocks and unsigned incidence matrices of complete graphs. Each incidence block has Smith form $1^{n-1}, 2$, and its order-two class

is exactly a divided-power vector $\theta^{[2]} \wedge z$. Third, the difference map of the Fano surface produces lifts of all ten quotient directions. The identity

$$\xi \star \theta^{[3]} = \pm \theta^{[2]} \wedge y(\xi)$$

for the Pontryagin product identifies their pushforwards and yields (3). No finite Smith-form computation enters the proof.

Beauville established the singularity and degree-six Fano geometry used here [Bea82]; Clemens and Griffiths supply the integral cylinder isomorphism and the minimal class of the Fano surface [CG72]. Kraemer already computed the rational decomposition-theorem correction for this resolution [Krä16, Corollary 6]; Section 6 explains why that calculation does not determine the integral middle lattice. The integral Lefschetz filtration of Faulkner Valiente and Miller Eismeier is the nearest general algebraic framework [FVME25]. We give the short off-central specialization needed here because it exposes the ten complete-graph parity defects directly.

The stronger assertion that the full degree-six Fano transfer surjects onto $H^3(M, \mathbf{Z})$ is not needed and is not claimed. Only ten endpoint classes, whose images in $H^3(X, \mathbf{Z})$ form a basis, enter the argument.

2 The topology of the resolution

Set $U = \Theta \setminus \{0\} = M \setminus X$, and let K be the link of the singular point. Since the blow-up is an isomorphism away from X , a punctured neighborhood of the singular point is a punctured tubular neighborhood of X in M . Hence K is the unit-circle bundle of $N_{X/M} = \mathcal{O}_X(-1)$.

Lemma 2.1. *The integral cohomology of the link in degrees at most four is*

$$H^1(K) = H^2(K) = 0, \quad H^3(K) \cong H^3(X) \cong \mathbf{Z}^{10}, \quad H^4(K) \cong \mathbf{Z}^{10} \oplus \mathbf{Z}/3.$$

The isomorphism in degree three is pullback along $K \rightarrow X$.

Proof. Let $h \in H^2(X, \mathbf{Z})$ be the hyperplane class and let $\ell \in H^4(X, \mathbf{Z})$ be the class of a line. The integral cohomology of a smooth cubic threefold is

$$H^k(X, \mathbf{Z}) = \mathbf{Z}, 0, \mathbf{Z}h, \mathbf{Z}^{10}, \mathbf{Z}\ell, 0, \mathbf{Z} \quad (0 \leq k \leq 6),$$

and multiplication by h sends 1 to h , h to 3ℓ , and ℓ to the point class. Apply the Gysin sequence of the circle bundle, whose Euler class is $-h$. The asserted groups and the degree-three pullback isomorphism follow at once. $\square$

The singular weak Lefschetz theorem for the support of an ample divisor gives

$$H^3(\Theta, \mathbf{Z}) \cong H^3(J, \mathbf{Z}) = \bigwedge^3 \Lambda; \tag{5}$$

one may use the relative homotopy form of stratified Morse theory [GM88, Part II, §1.2].

Proposition 2.2. *The maps b^* and e^* give the exact sequence (1).*

Proof. Excision at the isolated singularity identifies $H^k(\Theta, U; \mathbf{Z})$ with $\tilde{H}^{k-1}(K, \mathbf{Z})$. Since $H^2(K) = 0$, the pair sequence gives

$$0 \longrightarrow H^3(\Theta, \mathbf{Z}) \longrightarrow H^3(U, \mathbf{Z}) \longrightarrow H^3(K, \mathbf{Z}). \tag{6}$$

Now cover M by U and a tubular neighborhood of X . Their intersection retracts to K , and Mayer–Vietoris begins

$$0 \longrightarrow H^3(M, \mathbf{Z}) \longrightarrow H^3(U, \mathbf{Z}) \oplus H^3(X, \mathbf{Z}) \longrightarrow H^3(K, \mathbf{Z}).$$

The restriction $H^3(X, \mathbf{Z}) \rightarrow H^3(K, \mathbf{Z})$ is an isomorphism by Lemma 2.1. Projection therefore identifies $H^3(M, \mathbf{Z})$ with $H^3(U, \mathbf{Z})$. Under this identification the kernel of e^* is the image of $H^3(\Theta, \mathbf{Z})$, hence is $b^* \wedge^3 \Lambda$ by (5).

Surjectivity of e^* will follow in Proposition 4.3 from the universal-line correspondence. Since its target is free, the resulting exact sequence splits abstractly. It also proves that $H^3(M, \mathbf{Z})$ is free of rank $120 + 10 = 130$. $\square$

The argument deliberately postpones surjectivity. Local topology reduces it to the survival of all link classes; the Fano correspondence supplies integral lifts and fixes their global position in J at the same time.

3 The integral Lefschetz defect

We first work with a unimodular symplectic lattice Λ of rank $2g$, $g \geq 4$, and polarization θ . This isolates the only integral linear algebra in the paper.

Theorem 3.1. *For*

$$L_3 : \bigwedge^3 \Lambda \longrightarrow \bigwedge^5 \Lambda, \quad \alpha \longmapsto \theta \wedge \alpha,$$

one has

$$\text{Sat}(\text{im } L_3) = \text{im } L_3 + \theta^{[2]} \wedge \Lambda. \quad (7)$$

Moreover

$$\tau : \Lambda/2\Lambda \xrightarrow{\sim} \text{Sat}(\text{im } L_3)/\text{im } L_3, \quad z \longmapsto [\theta^{[2]} \wedge z] \quad (8)$$

is symplectically natural. The nonzero Smith factors of L_3 are all one except for exactly $2g$ factors equal to two.

Proof. Choose a symplectic basis $e_1, f_1, \dots, e_g, f_g$, and put $x_i = e_i \wedge f_i$. Label an exterior monomial by the symplectic pairs that occur as singletons and by those that occur as full pairs. Fix a singleton set A , choose e_i or f_i for each $i \in A$, and write u_A for their product. For $T \cap A = \emptyset$, put $x_T = \prod_{i \in T} x_i$. Multiplication by $\theta = \sum_i x_i$ preserves the singleton label and acts by

$$u_A x_T \longmapsto \sum_{j \notin A \cup T} u_A x_{T \cup \{j\}}. \quad (9)$$

These are integral direct-sum blocks; no rational primitive decomposition is being used.

In degree three there are two cases. If $|A| = 3$, then T is empty, and (9) sends one basis vector to an all-ones vector. Its image is primitive. If $|A| = 1$, then $|T| = 1$, and the block is the unsigned vertex–edge incidence map of the complete graph on $n = g - 1$ vertices:

$$U_n : \mathbf{Z}^n \longrightarrow \mathbf{Z}^{\binom{n}{2}}, \quad [i] \longmapsto \sum_{j \neq i} [\{i, j\}].$$

For $n \geq 3$, U_n is injective: the equations $a_i + a_j = 0$ on a triangle force every a_i to vanish. An $(n - 1)$ -minor is one, using the rows $\{1, j\}$, $2 \leq j \leq n$, and columns $2, \dots, n$. Every n -minor is even because modulo two the sum of all columns is zero. Finally, the rows of one triangle together with $\{1, k\}$, $4 \leq k \leq n$, give an n -minor of determinant ± 2 . Hence

$$\text{SNF}(U_n) = \text{diag}(1^{n-1}, 2). \quad (10)$$

The order-two class is represented by the sum of all edge vectors: twice that vector is the sum of all vertex columns, while the triangle equations show that it is not itself in the image.

There is one such block for each of the $2g$ choices of a singleton basis vector. Since

$$\theta^{[2]} = \sum_{i < j} x_i \wedge x_j,$$

the vector $\theta^{[2]} \wedge e_k$, or $\theta^{[2]} \wedge f_k$, is precisely the all-edge vector in the corresponding block. The blocks are independent, so these $2g$ vectors exhaust the saturation defect and prove (7) and (8).

Although the calculation used a basis, τ is defined by the polarization and its integral divided square. It therefore commutes with every element of $\mathrm{Sp}(\Lambda, \theta)$. $\square$

For $g = 5$, the domain has rank $\binom{10}{3} = 120$. Theorem 3.1 gives ten factors of two and 110 factors of one, proving the algebraic part of (2). This is an off-central integral Lefschetz map. The general filtration of [FVME25] provides a broader representation-theoretic setting; the occupancy proof above identifies the exact elementary divisors and their divided-power representatives in the degree required here.

4 The Fano surface and the geometric glue

Let F be the Fano surface of lines on X , and let $a : F \hookrightarrow J$ be its Albanese embedding. The difference map

$$\psi : F \times F \longrightarrow \Theta, \quad (r, s) \longmapsto a(s) - a(r)$$

is generically finite of degree six. Blowing up the diagonal gives a commutative diagram

$$\begin{array}{ccc} Y = \mathrm{Bl}_\Delta(F \times F) & \xrightarrow{q} & M \\ \downarrow \mu & & \downarrow b \\ F \times F & \xrightarrow{\psi} & J. \end{array}$$

The exceptional divisor of μ is $P = \mathbf{P}(T_F)$, the universal line $\{(\ell, x) : x \in \ell\} \subset F \times X$. Write $\pi : P \rightarrow F$ and $p : P \rightarrow X$ for its projections. These facts, as well as $[F] = \theta^{[3]}$, are classical [CG72, Bea82].

The cylinder map

$$\pi_* p^* : H^3(X, \mathbf{Z}) \longrightarrow H^1(F, \mathbf{Z})$$

is an integral isomorphism by Clemens–Griffiths [CG72, §2 and §11]. Its adjoint for the unimodular pairings is therefore an isomorphism

$$p_* \pi^* : H^3(F, \mathbf{Z}) / \mathrm{tors} \xrightarrow{\sim} H^3(X, \mathbf{Z}). \quad (11)$$

Lemma 4.1 (Clean exceptional restriction). *For $T \in H^3(F \times F, \mathbf{Z})$,*

$$e^* q_* \mu^* T = p_* \pi^* \Delta^* T, \quad b_* q_* \mu^* T = \psi_* T. \quad (12)$$

Proof. The equality $\psi^{-1}(0) = \Delta$ follows from injectivity of the Albanese embedding. Thus $q^{-1}(X) = P$ set-theoretically. It has multiplicity one: on a fibre C of $P \rightarrow F$, both q^*X and the exceptional divisor have degree -1 . Hence $q^*X = P$ as Cartier divisors and the normal bundles agree. Naturality of Thom classes gives $q^*e_* = j_*p^*$, where $j : P \hookrightarrow Y$. The first identity of (12) follows by adjunction and the projection formula; the second follows from $bq = \psi\mu$ and $\mu_*\mu^* = 1$. $\square$

We next calculate the endpoint of the difference correspondence. The Pontryagin product on J is denoted by $\star$.

Lemma 4.2 (Pontryagin endpoint). *For a principally polarized abelian fivefold and $\xi \in \bigwedge^9 \Lambda$, let $y(\xi) \in \Lambda$ be its integral symplectic wedge-dual. With one global sign depending on orientation and Pontryagin conventions,*

$$\xi \star \theta^{[3]} = \pm \theta^{[2]} \wedge y(\xi). \quad (13)$$

Proof. Choose a symplectic basis. It suffices to take ξ to be the oriented top monomial with one basis vector omitted. In the adjunction formula

$$\int_J (\xi \star \theta^{[3]}) \wedge \eta = \int_{J \times J} \text{pr}_1^* \xi \wedge \text{pr}_2^* \theta^{[3]} \wedge m^* \eta,$$

the first factor must receive the omitted vector, while the second receives the two complementary full symplectic pairs. Each admissible term occurs once because $\theta^{[3]}$ is a divided power. The result is the pairing of $\pm \theta^{[2]} \wedge y(\xi)$ with η . The degree-five wedge pairing is unimodular, which proves the identity. $\square$

For $\beta \in H^3(F, \mathbf{Z})/\text{tors}$, put $\xi = a_* \beta \in H^9(J, \mathbf{Z})$. The Albanese isomorphism, Poincare duality, and the principal polarization identify $\beta \mapsto y(\xi)$ with an integral isomorphism from the free part of $H^3(F, \mathbf{Z})$ to Λ . Combining it with the inverse of (11), and absorbing the global sign, defines the isomorphism $y : H^3(X, \mathbf{Z}) \rightarrow \Lambda$ used in the introduction.

Proposition 4.3 (Endpoint lifts). *For every $\gamma \in H^3(X, \mathbf{Z})$, there is $u_\gamma \in H^3(M, \mathbf{Z})$ such that*

$$e^* u_\gamma = \gamma, \quad b_* u_\gamma = \theta^{[2]} \wedge y(\gamma). \quad (14)$$

The common choice of sign is included in the definition of y .

Proof. Choose β with $p_* \pi^* \beta = \gamma$, and take $u_\gamma = q_* \mu^*(\beta \otimes 1)$, exchanging the two factors if needed to match the sign convention. Lemma 4.1 gives the restriction. Since the difference map is addition composed with $(-a) \times a$, the pushforward is

$$\psi_*(\beta \otimes 1) = (-1)^*(a_* \beta) \star [F].$$

Now $[F] = \theta^{[3]}$, and Lemma 4.2 gives the second identity. $\square$

Proposition 4.3 completes the proof of surjectivity in Proposition 2.2. Choose a basis of $H^3(X, \mathbf{Z})$ and the corresponding lifts. Together with $b^* \bigwedge^3 \Lambda$, they form a basis of $H^3(M, \mathbf{Z})$. Their pushforwards are, respectively, $\theta^{[2]} \wedge \Lambda$ and

$$b_* b^* \alpha = \theta \wedge \alpha.$$

Theorem 3.1 therefore proves (2).

Finally, (b_*, e^*) is injective. Indeed, on $\ker e^* = \text{im } b^*$, its first component is L , and L is injective by the block calculation in Section 3. Changing a lift of γ adds $b^* \alpha$, hence changes its pushforward by $L\alpha$. Equation (14) and the isomorphism (8) now give exactly the fibre product (3). This proves Theorem 1.1.

5 The dual escape lattice

We prove Theorem 1.2. Poincaré duality identifies $H^5(M, \mathbf{Z})/\text{tors}$ unimodularly with $\text{Hom}(H^3(M, \mathbf{Z}), \mathbf{Z})$. Under this identification, $b_* : H^5(M, \mathbf{Z})/\text{tors} \rightarrow H^7(J, \mathbf{Z})$ is restriction of functionals along $b^* : \bigwedge^3 \Lambda \rightarrow H^3(M, \mathbf{Z})$, followed by the wedge duality $(\bigwedge^3 \Lambda)^\vee \cong \bigwedge^7 \Lambda$. Since $b^* \bigwedge^3 \Lambda$ is a direct summand,

$$b_* : H^5(M, \mathbf{Z})/\text{tors} \longrightarrow \bigwedge^7 \Lambda \quad (15)$$

is surjective with primitive rank-ten kernel.

The exceptional Gysin map corresponds to

$$\beta \longmapsto (u \longmapsto \langle \beta, e^* u \rangle_X).$$

Surjectivity of e^* and unimodularity of the middle pairing on X identify its image with the kernel of (15).

It remains to quotient by pullbacks from J . For $w \in H^5(J, \mathbf{Z}) = \bigwedge^5 \Lambda$, the functional defined by $b^* w$ is

$$u \longmapsto \langle w, b_* u \rangle_J.$$

The lattice $S = b_* H^3(M, \mathbf{Z})$ is saturated in $\bigwedge^5 \Lambda$, so unimodularity of the middle wedge pairing shows that $\bigwedge^5 \Lambda \rightarrow S^\vee$ is surjective. Therefore

$$E_M \cong \text{Hom}(\ker(b_* : H^3(M, \mathbf{Z}) \rightarrow S), \mathbf{Z}). \quad (16)$$

In the fibre-product coordinates (3),

$$\ker b_* = \{(0, \gamma) : y(\gamma) \equiv 0 \pmod{2}\}.$$

Since y is unimodular, restriction identifies this kernel with $2H^3(X, \mathbf{Z})$. Thus (16) is free of rank ten. An exceptional Gysin functional pairs against $2H^3(X, \mathbf{Z})$, so its image in (16) is exactly twice the unimodular dual lattice. This is $2E_M$, proving Theorem 1.2.

Remark 5.1. The group $(\mathbf{Z}/2)^{10}$ occurs twice for dual reasons. In degree three it is the saturation quotient of the image in J . In degree five it is the quotient of a free rank-ten lattice by the exceptional sublattice. The escape lattice itself is not torsion.

6 Intersection cohomology of the theta divisor

For an isolated singularity in a complex fourfold, the middle-perversity Deligne sheaf agrees below the middle degree with the cohomology of the smooth locus. Thus

$$IH^3(\Theta, \mathbf{Z}) \cong H^3(U, \mathbf{Z}). \quad (17)$$

The Mayer–Vietoris proof of Proposition 2.2 identifies $H^3(M, \mathbf{Z}) \rightarrow H^3(U, \mathbf{Z})$ with an isomorphism. This proves the first assertion of Corollary 1.3. Under that identification, the pair sequence (6) and the surjectivity supplied by Proposition 4.3 give

$$0 \longrightarrow H^3(\Theta, \mathbf{Z}) \longrightarrow IH^3(\Theta, \mathbf{Z}) \longrightarrow H^3(X, \mathbf{Z}) \longrightarrow 0,$$

which proves the corollary.

For comparison, the rational decomposition of this resolution is already known. Kraemer proves, with complex coefficients, [Krä16, Corollary 6]

$$R\sigma_* \mathbf{Q}_M[4] \cong \text{IC}_\Theta \oplus \mathbf{Q}_0[2] \oplus \mathbf{Q}_0 \oplus \mathbf{Q}_0[-2], \quad (18)$$

where the same argument works over $\mathbf{Q}$. Equivalently,

$$H^k(M, \mathbf{Q}) \cong IH^k(\Theta, \mathbf{Q}) \oplus \begin{cases} \mathbf{Q}, & k = 2, 4, 6, \\ 0, & \text{otherwise.} \end{cases}$$

The three skyscraper summands come from the algebraic cohomology of the exceptional cubic. Equation (18) does not split the integral sequence in a manner compatible with (b_*, c^*) , nor does it determine the mod-two coset in (3).

The integral calculation outside degree three has an additional coefficient issue: in the link Gysin sequence, multiplication by the hyperplane class sends h to 3ℓ . We make no integral decomposition-theorem claim in those degrees. The conclusion here is exactly that the rank-130 lattice of Theorem 1.1 is the intrinsic middle intersection cohomology of Θ , while its rational even-degree correction is the prior result (18).

AI assistance disclosure

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